Process Management-I

(Introduction, Process, Threads & CPU Scheduling)

hich scheduling policy is most suitable for a me-shared operating systems?

a) Shortest Job First
) Round Robin
) First come first serve
) Elevator

[1995 : 1 M]

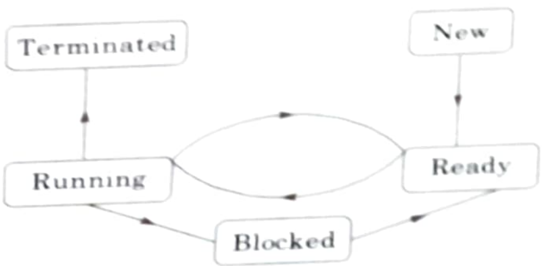
he sequence ..... is an optimal non-preemptive cheduling sequence for the following jobs which eaves the CPU idle for ..... unit(s) of time.

Job	Arrival time	Burst time
1	0.0	9
2	0.6	5
3	1.0	1

) {3, 2, 1}, 1 (b) {2, 1, 3}, 0
) {3, 2, 1}, 0 (d) {1, 2, 3}, 5

[1995 : 2 M]

he process state transition diagram in figure is epresentative of



) a batch operating system
) an operating system with a preemptive scheduler
) an operating system with a non-preemptive scheduler
) a uni-programmed operating system

[1996 : 1 M]

hich of the following is an example of spooled vice?

A line printer used to print the output of a number of jobs
A terminal used to enter input data to a running program
A secondary storage device in a virtual memory system
A graphic display device

[1996 : 1 M]

1.5 Four jobs to be executed on a single processor system arrive at time 0⁺ in the order A, B, C, D. Their burst CPU time requirements are 4, 1, 8, 1 time units respectively. The completion time of A under round robin scheduling with time slice of one time unit is

(a) 10 (b) 4
(c) 8 (d) 9

[1996 : 2 M]

1.6 Which of the following is an example of a spooled device?

(a) The terminal used to enter the input data for the C program being executed
(b) An output device used to print the output of a number of jobs.
(c) The secondary memory device in a virtual storage system
(d) The swapping area on a disk used by the swapper.

[1998 : 1 M]

1.7 Consider n processes sharing the CPU in a round robin fashion. Assuming that each process switch takes s seconds, what must be the quantum size q such that the overhead resulting from process switching is minimized but at the same time each process is guaranteed to get its turn at the CPU at least every t seconds?

(a) $q \leq \frac{t - ns}{n - 1}$ (b) $q \geq \frac{t - ns}{n - 1}$

(c) $q \leq \frac{t - ns}{n + 1}$ (d) $q \geq \frac{t - ns}{n + 1}$

[1998 : 1 M]

1.8 Four jobs are waiting to be run. Their expected run times are 6, 3, 5 and x in what order should they be run to minimize the average response time?

[1998 : 2 M]

1.9 System calls are usually invoked by using

(a) a software interrupt
(b) polling
(c) an indirect jump
(d) a privileged instruction

[1999 : 1 M]

1.10 A multi-user, multi-processing operating system cannot be implemented on hardware that does not support

(a) Address translation
(b) DMA for disk transfer
(c) At least two modes of CPU execution (privileged and non-privileged)
(d) Demand paging **[1999 : 2 M]**

1.11 Which of the following actions is/are typically not performed by the operating system when switching context from process A to process B?
(a) Saving current register values and restoring saved register values for process B.
(b) Changing address translation tables.
(c) Swapping out the memory image of process A to the disk.
(d) Invalidating the translation look-aside buffer. **[1999 : 2 M]**

1.12 A processor needs software interrupt to
(a) Test the interrupt system of the processor
(b) Implement co-routines
(c) Obtain system services which need execution of privileged instructions
(d) Return from subroutine **[2001 : 1 M]**

1.13 A CPU has two modes-privileged and non-privileged. In order to change the mode from privileged to non-privileged
(a) a hardware interrupt is needed
(b) a software interrupt is needed
(c) a privileged instruction (which does not generate an interrupt) is needed
(d) a non-privileged instruction (which does not generate an interrupt) is needed **[2001 : 1 M]**

1.14 Consider a set of n tasks with known runtimes $r_1, r_2, \dots, r_n$ to be run on a uniprocessor machine. Which of the following processor scheduling algorithms will result in the maximum throughput?
(a) Round-Robin
(b) Shortest-Job-First
(c) Highest-Response-Ratio-Next
(d) First-Come-First-Served **[2001 : 1 M]**

1.15 Which of the following scheduling algorithms is non-preemptive?
(a) Round Robin
(b) First-in-First-out
(c) Multilevel Queue Scheduling
(d) Multilevel Queue Scheduling with Feedback **[2001 : 1 M]**

1.16 Which of the following does not interrupt a running process?
(a) A device (b) Timer
(c) Scheduler process (d) Power failure **[2001 : 2 M]**

1.17 Which combination of the following features will suffice to characterize an OS as a multi-programmed OS? (i) more than one program may be loaded into main memory at the same time for execution. (ii) If a program waits for certain events such as I/O, another program is immediately scheduled for execution. (iii) If the execution of a program terminates, another program is immediately scheduled for execution
(a) (i) (b) (i) and (ii)
(c) (i) and (iii) (d) (i), (ii) and (iii) **[2002 : 2 M]**

1.18 Draw the process state transition diagram of an OS in which (i) each process is in one of the five states: created, ready, running, blocked (i.e. sleep or wait), or terminated, and (ii) only non-preemptive scheduling is used by the OS. Label the transitions appropriately. **[2002 : 2 M]**

1.19 A uni-processor computer system only has two processes, both of which alternate 10 ms CPU bursts with 90 ms I/O bursts. Both the processes were created at nearly the same time. The I/O of both processes can proceed in parallel. Which of the following scheduling strategies will result in the least CPU utilization (over a long period of time) for this system?
(a) First come first served scheduling
(b) Shortest remaining time first scheduling
(c) Static priority scheduling with different priorities for the two processes
(d) Round robin scheduling with a time quantum of 5 ms. **[2003 : 2 M]**

1.20 Consider the following statements with respect to user-level threads and kernel-supported threads
(i) Context switch is faster with Kernel-supported threads
(ii) For user-level threads, a system call can block the entire process
(iii) Kernel-supported threads can be scheduled independently
(iv) User-level threads are transparent to the Kernel
Which of the above statements are true?
(a) (ii), (iii) and (iv) only
(b) (ii) and (iii) only
(c) (i), and (iii) only
(d) (i) and (ii) only **[2004 : 1 M]**

1.21 Which one of the following is NOT shared by the threads of the same process?
(a) Stack
(b) Address Space
(c) File Descriptor Table
(d) Message Queue **[2004 : 1 M]**

1.22 A process executes the following segment of code:

```
for(i = 1; i < = n; i++)
fork( );
```

The number of new processes created is
(a) n (b) ((n(n + 1))/2)
(c) $2^n - 1$ (d) $3^n - 1$
[2004 : 2 M]

1.23 Consider the following set of processes, with the arrival times and the CPU-burst times given in milliseconds

Process	Arrival time	Burst time
P1	0	5
P2	1	3
P3	2	3
P4	4	1

What is the average turnaround time for these processes with the preemptive Shortest Remaining Processing Time first (SRPT) algorithm?
(a) 5.50 (b) 5.75
(c) 6.00 (d) 6.25 **[2004 : 2 M]**

1.24 We wish to schedule three processes P1, P2 and P3 on a uniprocessor system. The priorities, CPU time requirements and arrival times of the processes are as shown below:

Process	Priority	CPU time required	Arrival time (hh:mm:ss)
P1	10 (highest)	20 sec	00:00:05
P2	9	10 sec	00:00:03
P3	8 (lowest)	15 sec	00:00:00

We have a choice of preemptive or non-preemptive scheduling. In preemptive scheduling, a late-arriving higher priority process can preempt a currently running process with lower priority. In non-preemptive scheduling, a late-arriving higher priority process must wait for the currently executing process to complete before it can be scheduled on the processor. What are the turnaround times (time from arrival till completion) of P2 using preemptive and non-preemptive scheduling respectively?

(a) 30 sec, 30 sec (b) 30 sec, 10 sec
(c) 42 sec, 42 sec (d) 30 sec, 42 sec
[2005 : 2 M]

1.25 Two shared resources R_1 and R_2 are used by processes P_1 and P_2. Each process has a certain priority for accessing each resource. Let T_{ij} denote the priority of P_i for accessing R_j. A process P_i can snatch a resource R_k from process P_j if T_{ik} is greater than T_{jk}.
Given the following :
I. $T_{11} > T_{21}$ II. $T_{12} > T_{22}$
III. $T_{11} < T_{21}$ IV. $T_{12} < T_{22}$
Which of the following conditions ensures that P_1 and P_2 can never deadlock?
(a) (I) and (IV) (b) (II) and (III)
(c) (I) and (II) (d) None of the above
[2005 : 2 M]

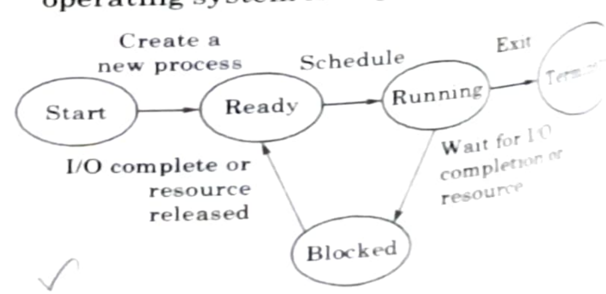
1.26 Consider the following code fragment:

```
if (fork ( ) = = 0)
{a = a + 5; printf("%d, %d\n", a, &a); }
else{a = a – 5; printf("%d, %d\n", a, &a); }
```

Let u, v be the values printed by the parent process, and x, y be the values printed by the child process. Which one of the following is TRUE?
(a) u = x + 10 and v = y
(b) u = x + 10 and v ≠ y
(c) u + 10 = x and v = y
(d) u + 10 = x and v ≠ y **[2005 : 2 M]**

1.27 Consider three CPU-intensive processes, which require 10, 20 and 30 time units and arrive at times 0, 2, and 6, respectively. How many context switches are needed if the operating system implements a shortest remaining time first scheduling algorithm? Do not count the context switches at time zero and at the end
(a) 1 (b) 2
(c) 3 (d) 4 **[2006 : 1 M]**

1.28 The process state transition diagram of an operating system is as given below.



Which of the following must be FALSE about the above operating system?
(a) It is a multiprogrammed operating system
(b) It uses preemptive scheduling
(c) It uses non-preemptive scheduling
(d) It is a multi-user operating system
[2006 : 1 M]

1.29 The arrival time, priority, and durations of the CPU and I/O bursts for each of three processes P_1, P_2 and P_3 are given in the table below. Each process has a CPU burst followed by an I/O burst followed by another CPU burst. Assume that each process has its own I/O resource.

Process	Arrival time	Priority	Burst durations CPU, I/O CPU
P_1	0	2	1, 5, 3
P_2	2	3 (lowest)	3, 3, 1
P_3	3	1 (highest)	2, 3, 1

The multi-programmed operating system uses preemptive priority scheduling. What are the finish times of the processes P_1, P_2 and P_3?
(a) 11, 15, 9
(b) 10, 15, 9
(c) 11, 16, 10
(d) 12, 17, 11
[2006 : 2 M]

1.30 Consider three processes, all arriving at time zero, with total execution time of 10, 20 and 30 units, respectively. Each process spends the first 20% of execution time doing I/O, the next 70% of time doing computation, and the last 10% of time doing I/O again. The operating system uses a shortest remaining compute time first scheduling algorithm and schedules a new process either when the running process get blocked on I/O or when the running process finishes its compute burst. Assume that all I/O operations can be overlapped as much as possible. For what percentage of time does the CPU remain idle?
(a) 0%
(b) 10.6%
(c) 30.0%
(d) 89.4%
[2006 : 2 M]

1.31 Consider three processes (process id 0, 1, 2, respectively) with compute time bursts 2, 4, and 8 time units. All processes arrive at time zero. Consider the longest remaining time first (LRTF) scheduling algorithm. In LRTF ties are broken by giving priority to the process with the lowest process id. The average turn around time is
(a) 13 units
(b) 14 units
(c) 15 units
(d) 16 units **[2006 : 2 M]**

1.32 **List-I** contains some CPU scheduling algorithms and **List-II** contains some applications. Match entries in **List-I** entries in **List-II**

List-I
A. Gang Scheduling
B. Rate Monotonic Scheduling
C. Fair Share Scheduling

List-II
1. Guaranteed Scheduling
2. Real-time Scheduling
3. Thread Scheduling

Code:

	A	B	C
(a)	3	2	1
(b)	1	2	3
(c)	2	3	1
(d)	1	3	2

[2007 : 1 M]

1.33 Consider the following statements about user level threads and kernel level threads. Which one of the following statements is FALSE?
(a) Context switch time is longer for Kernel level threads than for user level threads
(b) User level threads do not need any hardware support
(c) Related Kernel level threads can be scheduled on different processors in a multiprocessor system
(d) Blocking one Kernel level thread blocks all related threads **[2007 : 1 M]**

1.34 An operating system uses Shortest Remaining Time first (SRT) process scheduling algorithm. Consider the arrival times and execution times for the following processes

Process	Execution time	Arrival time
P1	20	0
P2	25	15
P3	10	30
P4	15	45

What is the total waiting time for process P2?
(a) 5
(b) 15
(c) 40
(d) 55 **[2007 : 2 M]**

1.35 If the time-slice used in the round-robin scheduling policy is more than the maximum time required to execute any process, then the policy will
(a) degenerate to shortest job first
(b) degenerate to priority scheduling
(c) degenerate to first come first serve
(d) None of the above **[2008 : 2 M]**

1.36 Which of the following statements about synchronous and asynchronous I/O is NOT true?

(a) An ISR is invoked on completion of I/O in synchronous I/O but not in asynchronous I/O.
(b) In both synchronous and asynchronous I/O an ISR (Interrupt Service Routine) is invoked after completion of the I/O.
(c) A process making a synchronous I/O call waits until I/O is complete, but a process making an asynchronous I/O call does not wait for completion of the I/O.
(d) In the case of synchronous I/O, the process waiting for the completion of I/O is woken up by the ISR that is invoked after the completion of I/O.

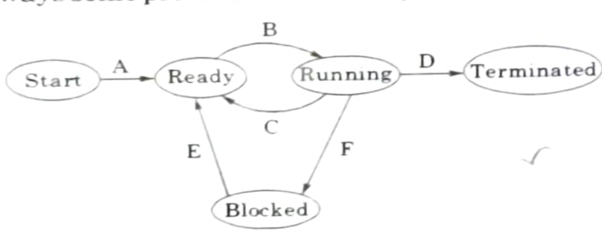
[2008 : 2 M]

1.37 A process executes the following code for (i = 0; i < n; i ++) fork ( );
The total number of child processes created is

(a) n (b) $2^n - 1$
(c) 2^n (d) $2^{n+1} - 1$

[2008 : 2 M]

1.38 In the following process state transition diagram for a uniprocessor system, assume that there are always some processes in the ready state:



Now consider the following statements:

I. If a process makes a transition D, it would result in another process making transition A immediately.
II. A process P_2 in blocked state can make transition E while another process P_1 is in running state.
III. The OS uses preemptive scheduling.
IV. The OS uses non-preemptive scheduling.

Which of the above statements are TRUE?

(a) I and II (b) I and III
(c) II and III (d) II and IV

[2009 : 2 M]

1.39 Which of the following statements are true?

I. Shortest remaining time first scheduling may cause starvation.
II. Preemptive scheduling may cause starvation.
III. Round robin is better than FCFS in terms of response time.

(a) I only (b) I and III only
(c) II and III only (d) I, II and III

[2010 : 1 M]

1.40 Let the time taken to switch between user and Kernel modes of execution be t_1 while the time taken to switch between two processes be t_2. Which of the following is TRUE?

(a) $t_1 > t_2$
(b) $t_1 = t_2$
(c) $t_1 < t_2$
(d) nothing can be said about the relation between t_1 and t_2

[2011 : 1 M]

1.41 A computer handles several interrupt sources of which the following are relevant for this question.

- Interrupt from CPU temperature sensor (raises interrupt if CPU temperature is too high)
- Interrupt from Mouse (raises interrupt if the mouse is moved or a button is pressed)
- Interrupt from Keyboard (raises interrupt when a key is pressed or released)
- Interrupt from Hard Disk (raises interrupt when a disk read is completed)

Which one of these will be handled at the **HIGHEST** priority?

(a) Interrupt from Hard Disk
(b) Interrupt from Mouse
(c) Interrupt from Keyboard
(d) Interrupt from CPU temperature sensor

[2011 : 1 M]

1.42 A thread is usually defined as a "light weight process" because an Operating System (OS) maintains smaller data structures for a thread than for a process. In relation to this, which of the following is TRUE?

(a) On per-thread basis, the OS maintains only CPU register state
(b) The OS does not maintain a separate stack for each thread
(c) On per-thread basis, the OS does not maintain virtual memory state
(d) On per-thread basis, the OS maintains only scheduling and accounting information

[2011 : 1 M]

1.43 Consider the following table of arrival time and burst time for three processes P0, P1 and P2.

Process	Arrival time	Burst time
P0	0 ms	9 ms
P1	1 ms	4 ms
P2	2 ms	9 ms

The pre-emptive shortest job first scheduling algorithm is used. Scheduling is carried out only at arrival or completion of processes. What is the average waiting time for the three processes?

(a) 5.0 ms
(b) 4.33 ms
(c) 6.33 ms
(d) 7.33 ms

[2011 : 2 M]

1.44 A process executes the code:

```
fork( );
fork( );
fork( );
```

The total number of child processes created is

(a) 3
(b) 4
(c) 7
(d) 8

[2012 : 1 M]

1.45 Consider the 3 processes, P1, P2 and P3 shown in the table:

Process Name	Arrival Time	Time Unit Required
P1	0	5
P2	1	7
P3	3	4

The completion order of the 3 processes under the policies FCFS and RR2 (round robin scheduling with CPU quantum of 2 time units) are:

(a) **FCFS:** P1, P2, P3 **RR2:** P1, P2, P3
(b) **FCFS:** P1, P3, P2 **RR2:** P1, P3, P2
(c) **FCFS:** P1, P2, P3 **RR2:** P1, P3, P2
(d) **FCFS:** P1, P3, P2 **RR2:** P1, P2, P3

[2012 : 2 M]

1.46 A scheduling algorithm assigns priority proportional to the waiting time of a process. Every process starts with priority zero (the lowest priority). The schedule reevaluates the process priorities every T time units and decides the next process to schedule. Which one of the following is TRUE if the processes have no I/O operations and all arrive at time zero?

(a) This algorithm is equivalent to the first come first serve algorithm
(b) This algorithm is equivalent to the round-round algorithm
(c) This algorithm is equivalent to the shortest-job-first algorithm
(d) This algorithm is equivalent to the shortest-remaining-time-first algorithm

[2013 : 1 M]

1.47 Which one of the following is **FALSE**?

(a) User level threads are not scheduled by the Kernel.
(b) When a user level thread is blocked, all other threads of its process are blocked.
(c) Context switching between user level threads is faster than context switching between Kernel level threads.
(d) Kernel level threads cannot share the code segment.

[2014 (Set-1) : 1 M]

1.48 Consider the following set of processes that need to be scheduled on a single CPU. All the times are given in milliseconds.

Process Name	Arrival Time	Execution Time
A	0	6
B	3	2
C	5	4
D	7	6
E	10	3

Using the shortest remaining time first scheduling algorithm, the average process turnaround time (in msec) is ________.

[2014 (Set-1) : 2 M]

1.49 Three processes A, B and C each execute a loop of 100 iterations. In each iteration of the loop, a process performs a single computation that requires t_c CPU milliseconds and then initiates a single I/O operation that lasts for t_{io} milliseconds. It is assumed that the computer where the processes execute has sufficient number of I/O devices and the OS of the computer assigns different I/O devices to each process. Also, the scheduling overhead of the OS is negligible. The processes have the following characteristics:

Process id	t_c	t_{io}
A	100 ms	500 ms
B	350 ms	500 ms
C	200 ms	500 ms

The processes A, B, and C are started at times 0, 5 and 10 milliseconds respectively, in a pure time sharing system (round robin scheduling) that uses a time slice of 50 milliseconds. The time in milliseconds at which process C would complete its first I/O operation is ________.

[2014 (Set-2) : 2 M]

1.50 An operating system uses shortest remaining time first scheduling algorithm for pre-emptive scheduling of processes. Consider the following set of processes with their arrival times and CPU burst times (in milliseconds):

Process	Arrival Time	Burst Time
P1	0	12
P2	2	4
P3	3	6
P4	8	5

The average waiting time (in milliseconds) of the processes is ________ **[2014 (Set-3) : 2 M]**

1.51 Consider a uniprocessor system executing three tasks T_1, T_2 and T_3, each of which is composed of an infinite sequence of jobs (or instances) which arrive periodically at intervals of 3, 7 and 20 milliseconds, respectively. The priority of each task is the inverse of its period, and the available tasks are scheduled in order of priority, with the highest priority task scheduled first. Each instance of T_1, T_2 and T_3 requires an execution time of 1, 2 and 4 milliseconds, respectively. Given that all tasks initially arrive at the beginning of the 1st millisecond and task preemptions are allowed, the first instance of T_3 completes its execution at the end of ________ milliseconds. **[2015 (Set-1) : 2 M]**

1.52 The maximum number of processes that can be in Ready state for a computer system with n CPUs is

(a) n (b) n^2
(c) 2^n (d) Independent of n

[2015 (Set-3) : 1 M]

1.53 For the processes listed in the following table, which of the following scheduling schemes will give the lowest average turnaround time?

Process	Arrival Time	Processing Time
A	0	3
B	1	6
C	4	4
D	6	2

(a) First Come First Serve
(b) Non-preemptive Shortest Job First
(c) Shortest Remaining Time
(d) Round-Robin with Quantum value two

[2015 (Set-3) : 2 M]

1.54 Consider an arbitrary set of CPU-bound processes with unequal CPU burst lengths submitted at the same time to a computer system. Which one of the following process scheduling algorithms would minimize the average waiting time in the ready queue?

(a) Shortest remaining time first
(b) Round-robin with time quantum less than the shortest CPU burst
(c) Uniform random
(d) Highest priority first with pr... proportional to CPU burst length

[2016 (Set-1) : 1 M]

1.55 Consider the following processes, with the ar... time and the length of the CPU burst giv... milliseconds. The scheduling algorithm us... preemptive shortest remaining-time first.

Process	Arrival Time	Burst Time
P_1	0	10
P_2	3	6
P_3	7	1
P_4	8	3

The average turn around time of these processes is ________ milliseconds.

[2016 (Set-2) : 2 M]

1.56 Threads of a process share

(a) global variables but not heap
(b) heap but not global variables
(c) neither global variables nor heap
(d) both heap and global variables

[2017 (Set-1) : 1 M]

1.57 Consider the following CPU processes wi... arrival times (in milliseconds) and length of CP... bursts (in milliseconds) as given below:

Process	Arrival time	Burst time
P_1	0	7
P_2	3	3
P_3	5	5
P_4	6	2

If the pre-emptive shortest remaining time fi... scheduling algorithm is used to schedule th... processes, then the average waiting time acr... all processes is ______ milliseconds.

[2017 (Set-1) : 1 M]

1.58 Which of the following is/are shared by all th... threads in a process?

I. Program counter II. Stack
III. Address space IV. Registers

(a) I and II only (b) III only
(c) IV only (d) III and IV only

[2017 (Set-2) : 1 M]

1.59 Consider the set of processes with arrival ti... (in milliseconds), CPU burst time ... milliseconds), and priority (0 is the high... priority) shown below. None of the processes h... I/O burst time.

Process	Arrival Time	Burst Time	Priority
[illegible]	[illegible]	[illegible]	[illegible]

The average waiting time (in milliseconds) of all the processes using preemptive priority scheduling algorithm is ____

[2017 (Set-2) : 2 M]

[illegible] Consider the following C program is executed on a Unix/Linux system:

```
#include<unistd.h>
int main( )
{
    int i;
    for (i = 0; i < 10; i++)
        if (i % 2 == 0) fork( );
    return 0;
}
```

The total number of child processes created is ____

[2019 : 1 M]

[illegible] Consider the following four processes with arrival times (in milliseconds) and their length of CPU burst (in milliseconds) as shown below:

Process	P_1	P_2	P_3	P_4
Arrival time	[illegible]	[illegible]	[illegible]	[illegible]
CPU time	[illegible]	[illegible]	[illegible]	[illegible]

These processes are run on a single processor using preemptive Shortest Remaining Time First scheduling algorithm. If the average waiting time of the processes is 1 millisecond, then the value of Z is ____

[2019 : 2 M]

1.62 Consider the following statements about process state transitions for a system using preemptive scheduling.

I. A running process can move to ready state.
II. A ready process can move to running state.
III. A blocked process can move to running state.
IV. A blocked process can move to ready state.

Which of the above statements are TRUE?

(a) I, II, III and IV
(b) II and III only
(c) I, II and III only
(d) I, II and IV only

[2020 : 1 M]

[illegible] Consider the following set of processes, assumed to have arrived at time 0. Consider the CPU scheduling algorithms Shortest Job First (SJF) and Round Robin (RR). For RR, assume that the processes are scheduled in the order [illegible]

Processes	[illegible]	[illegible]	[illegible]	[illegible]
Burst time (in ms)	[illegible]	[illegible]	[illegible]	[illegible]

If the time quantum for RR is [illegible] ms, then the absolute value of the difference between the average turnaround times (in ms) of SJF and RR (round off to 2 decimal places) is ____

[2020 : 1 M]

[illegible] Three processes arrive at time zero with CPU bursts of [illegible] and [illegible] [illegible]. If the scheduler has prior knowledge about the length of the CPU bursts, [illegible] average waiting time [illegible] in a non-preemptive [illegible] algorithm [illegible]

[2021 (Set-1) : 1 M]

[illegible] Which of the following standard C library functions will always invoke a system call when executed from a single-threaded process in a UNIX/Linux operating system?

(a) exit
(b) malloc
(c) sleep
(d) strlen

[2021 (Set-1) : 1 M]

[illegible] Which of the following statement(s) is/are correct in the context of CPU scheduling?

(a) [illegible] and minimize the [illegible]
(b) [illegible] waiting time
(c) [illegible]
(d) [illegible]

[2021 (Set-1) : 1 M]

[illegible]

considered a context switch. Which one of the following is NOT possible as CPU burst time (in time units) of these processes?

(a) P = 4, Q = 10, R = 6, S = 2
(b) P = 2, Q = 9, R = 5, S = 1
(c) P = 4, Q = 12, R = 5, S = 4
(d) P = 3, Q = 7, R = 7, S = 3

[2022 : 2 M]

1.68 Which one or more of the following need to be saved on a context switch from one thread (T1) of a process to another thread (T2) of the same process?

(a) Page table base register
(b) Stack pointer
(c) Program counter
(d) General purpose registers

[2023 : 1 M]

1.69 Which one or more of the following options guarantee that a computer system will transition from user mode to Kernel mode?

(a) Function call
(b) Malloc call
(c) Page fault
(d) System call

[2023 : 1 M]

1.70 Which one or more of the following CPU scheduling algorithms can potentially cause starvation?

(a) First-in First-Out
(b) Round Robin
(c) Priority Scheduling
(d) Shortest Job First

[2023 : 1 M]

1.71 Which of the following statements about threads is/are TRUE?

(a) Threads can only be implemented in Kernel space
(b) Each thread has its own file descriptor table for open files
(c) Threads belonging to a process are by default not protected from each other
(d) All the threads belonging to a process share a common stack

[2024 (Set-1) : 1 M]

1.72 Which of the following process state transitions is/are NOT possible?

(a) Waiting to Running
(b) Ready to Waiting
(c) Running to Terminated
(d) Running to Ready

[2024 (Set-1) : 1 M]

1.73 Consider the following two threads T1 and T2 that update two shared variables a and b. Assume that initially $a = b = 1$. Though context switching between threads can happen at any time, each statement of T1 or T2 is executed atomically without interruption.

T1	T2
$a = a + 1;$	$b = 2 * b;$
$b = b + 1;$	$a = 2 * a;$

Which one of the following options lists all the possible combinations of values of a and b after both T1 and T2 finish execution?

(a) $(a = 3, b = 4)$; $(a = 4, b = 3)$; $(a = 3, b = 3)$
(b) $(a = 4, b = 4)$; $(a = 4, b = 3)$; $(a = 3, b = 4)$
(c) $(a = 2, b = 2)$; $(a = 2, b = 3)$; $(a = 3, b = 4)$
(d) $(a = 4, b = 4)$; $(a = 3, b = 3)$; $(a = 4, b = 3)$

[2024 (Set-1) : 2 M]

1.74 Consider a process P running on a CPU. Which one or more of the following events will always trigger a context switch by the OS that results in process P moving to a non-running state (e.g., ready, blocked)?

(a) A timer interrupt is raised by the hardware
(b) P tries to access a page that is in the swap space, triggering a page fault.
(c) An interrupt is raised by the disk to deliver data requested by some other process.
(d) P makes a blocking system call to read a block of data from the disk.

[2024 (Set-2) : 1 M]

1.75 Consider a multi-threaded program with two threads T1 and T2. The threads share two semaphores: s1 (initialized to 1) and s2 (initialized to 0). The threads also share a global variable x (initialized to 0). The threads execute the code shown below.

```
// code of T1          // code of T2
wait (s1);             wait (s1);
x = x + 1;             x = x + 1;
print(x);              print(x);
wait(s2);              signal(s2);
signal(s1);            signal(s1);
```

Which of the following outcomes is/are possible when threads T1 and T2 execute concurrently?

(a) T2 runs first and prints 1, T1 runs next and prints 2

(b) T2 runs first and prints 1, T1 does not print anything (deadlock)

(c) T1 runs first and prints 1, T2 runs next and prints 2

(d) T1 runs first and prints 1, T2 does not print anything (deadlock)

[2024 (Set-2) : 2 M]

1.76 Consider a single processor system with four processes A, B, C, and D, represented as given below, where for each process the first value is its arrival time, and the second value is its CPU burst time.

A(0, 10), B(2, 6), C(4, 3) and D(6, 7)

Which one of the following options gives the average waiting times when preemptive Shortest Remaining Time First (SRTF) and Non-Preemptive Shortest Job First (NP-SJF) CPU scheduling algorithms are applied to the processes?

(a) SRTF = 6, NP-SJF = 7

(b) SRTF = 7, NP-SJF = 7.5

(c) SRTF = 7, NP-SJF = 8.5

(d) SRTF = 6, NP-SJF = 7.5

[2024 (Set-2) : 2 M]

■■■■

Answers Process Management-I

1.1	(b)	**1.2**	(a)	**1.3**	(b)	**1.4**	(a)	**1.5**	(d)	**1.6**	(b)	**1.7**	(a)	**1.8**	Sol.	**1.9**	(a)
1.10	(c)	**1.11**	(c)	**1.12**	(c)	**1.13**	(d)	**1.14**	(b)	**1.15**	(b)	**1.16**	(a)	**1.17**	(d)	**1.18**	Sol.
1.19	(d)	**1.20**	(a)	**1.21**	(a)	**1.22**	(c)	**1.23**	(a)	**1.24**	(d)	**1.25**	(c)	**1.26**	(d)	**1.27**	(b)
1.28	(b)	**1.29**	(b)	**1.30**	(b)	**1.31**	(a)	**1.32**	(a)	**1.33**	(d)	**1.34**	(b)	**1.35**	(c)	**1.36**	(b)
1.37	(b)	**1.38**	(c)	**1.39**	(d)	**1.40**	(c)	**1.41**	(d)	**1.42**	(c)	**1.43**	(a)	**1.44**	(c)	**1.45**	(c)
1.46	(b)	**1.47**	(d)	**1.48**	(7.2)	**1.49**	(1000)	**1.50**	(5.5)	**1.51**	(12)	**1.52**	(d)	**1.53**	(c)	**1.54**	(a)
1.55	(8.25)	**1.56**	(d)	**1.57**	(3)	**1.58**	(b)	**1.59**	(29)	**1.60**	(31)	**1.61**	(2)	**1.62**	(d)	**1.63**	(5.25)
1.64	(12)	**1.65**	(a, c)	**1.66**	(b, c, d)	**1.67**	(d)	**1.68**	(b, c, d)	**1.69**	(c, d)	**1.70**	(c, d)	**1.71**	(c)	**1.72**	(a, b)
1.73	(d)	**1.74**	(b, d)	**1.75**	(a, d)	**1.76**	(d)										